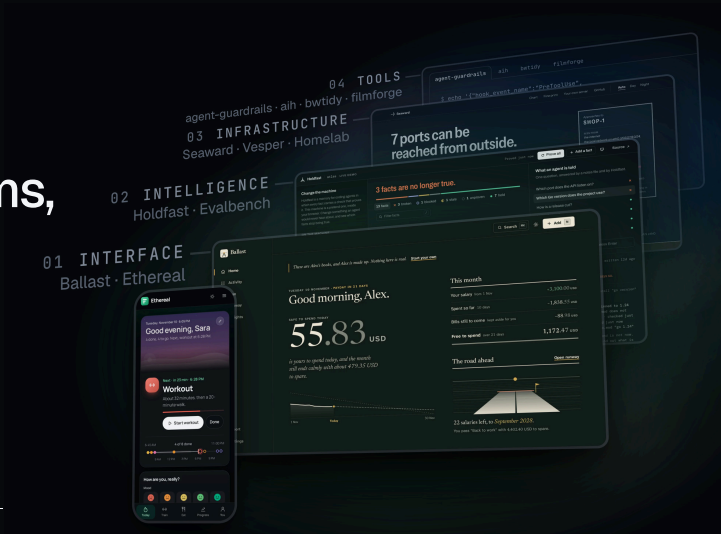
Ali Husain

Software engineer and product designer

I build complete systems, from the interface to the infrastructure.



ali@gallowforge.dev

gallowforge.github.io github.com/gallowforge

Remote, UTC+3 · Open to remote work

IN NUMBERS

11
public repositories, each one specified, tested, documented and released

4,503
automated tests across them, run on every change

15
self-hosted services in production on a platform I designed and operate

SELECTED WORK

Holdfast

COMMAND-LINE TOOL
OPEN SOURCE

A memory for coding agents in which every fact carries its own proof.
Each fact is stored with a check, run again before the agent is told. What no longer holds comes back as broken.

1,651 automated tests, on Linux and macOS
10 kinds of check, from a file to a live endpoint

gallowforge.github.io/holdfast

Seaward

COMMAND-LINE TOOL
OPEN SOURCE

Shows which ports on a server can really be reached from outside, and the rule that decides each one.
Follows a packet through the rules the kernel has loaded, so a port Docker published past the firewall's list is found.

1,212 automated tests, run on every change
2,111 real packets checked against the kernel

gallowforge.github.io/seaward

Ballast

LIVE APP
OPEN SOURCE

Personal finance that turns savings into a salary that lasts.
Designed and built end to end, offline-first. Says what is safe to spend today and reads bank statements on the device.

599 automated tests
40 currencies

gallowforge.github.io/ballast

Ethereal

LIVE APP
OPEN SOURCE

A calm coach for training, food and the shape of the day.
An installable app that plans the day, guides workouts and tracks nutrition. Designed, built and delivered end to end.

397 automated tests
1,474 foods, offline

gallowforge.github.io/Ethereal

Evalbench

SELF-HOSTED APP
OPEN SOURCE

Names the language model to use for your own job, and shows the proof.
Finds the cheapest model that clears your bar. Answers are graded by automatic checks, then rated blind in pairs.

291 automated tests, plus sandbox checks in CI
7 check types, up to sandboxed code tests

gallowforge.github.io/evalbench

Vesper

SELF-HOSTED APP
OPEN SOURCE

A read-only operations dashboard that answers one question: does anything need you?
One Go binary, read-only by construction. When it cannot confirm something it says so instead of showing an all-clear.

7.4 MB one Go binary, about 27 MB of memory
191 tests, 34 end to end under real systemd

gallowforge.github.io/vesper

Homelab

PLATFORM
OPEN SOURCE

Fifteen services on one small machine, run as code, with backups that are proven every week.
Zero port forwards. Every backup set is verified nightly, copied to three machines and restored in a weekly drill.

15 containers, 21 timers, 43 automated tests
77 alert conditions and 13 uptime probes

github.com/gallowforge/homelab

Ali Husain Software engineer and product designer ali@gallowforge.dev

CAPABILITIES		Where to see it
Product design	Interfaces designed for phone and desktop together and accessible by default, each with its own design system, light and dark themes, data charts and motion.	Ballast, Ethereal, Vesper Ethereal scores Lighthouse 100 on key pages, swept against WCAG 2.1 AA
Front end	Installable apps that work offline with no account or server, statements read on the device, Go engines in the browser as WebAssembly, and a live dashboard streamed to it.	Ballast, Ethereal, Vesper, Seaward 599 and 397 automated tests on the two apps
Back end	Services and command-line tools in Go, Python, TypeScript, Bash, SQL, C and C++, cross-compiled to single binaries and shipped with their documentation.	Holdfast, Seaward, Vesper, Evalbench Three Go binaries, each with no third-party Go dependencies
Data and statistics	SQLite schema design with full-text search, SQL triggers, statement parsing and reconciliation, exact money arithmetic, and the statistics behind a fair comparison.	Ballast, Evalbench, aih 40 currencies in exact integer arithmetic; BM25 full-text search
AI evaluation	Rubrics, automatic checks and sandboxed code tests, then blind pairwise rating scored with Bradley-Terry ratings and confidence intervals, and preference pairs for training.	Evalbench Finds the cheapest model that clears your bar; 7 check types
Agent engineering	Agent memory re-checked before every use, an agent tool server, hooks and skills, reviewer and verifier agents that judge work they did not write, and an always-on operations agent.	Holdfast, agent-guardrails, Homelab Over 400 agent runs; coordinated teams of 9, 13 and 98 agents
Infrastructure and operations	Linux, containers, systemd, CI/CD and infrastructure as code with drift checks, monitoring with a dead-man switch, and backups restored in a drill every week.	Homelab, Vesper 450 commits, 21 systemd timers, about 4,000 lines of runbooks
Networking	DNS for a whole house over HTTPS with DNSSEC, TLS at one reverse proxy, a mesh VPN through carrier-grade NAT with zero port forwards, and Linux firewall internals.	Seaward, Homelab 2,111 real packets checked against the kernel; 7-node mesh VPN
Security practice	Threat models, privilege separation, systemd and container sandboxing, password hashing, web hardening, secret scanning and vault audits; zero port forwards.	Seaward, Homelab, Vesper, Evalbench systemd exposure scores 0.5 and 2.4; code tests run with no network
Testing and releases	Unit, end-to-end, fuzz and golden tests, differential tests against the real Linux kernel, accessibility and performance gates in CI, and checksummed, versioned releases.	All 11 public repositories 4,503 tests, 36 releases, 21 CI workflows, 12 fuzz targets
Video and audio	Painted films made with AI image and video models and finished in code: a beat tracker written from scratch, broadcast loudness and 4K mastering on the GPU.	filmforge 9 skills and 7 tested tools; every encode verified
Smart home and media	Home automation across lights, locks, TV and audio in one home app, lighting that follows the TV picture, self-hosted media streaming, and room acoustics tuned by measurement.	Homelab 15 self-hosted services; ambient TV lighting write-up

HOW I WORK

Specified before it is built, tested before it ships, documented so it runs without me.

Prove it works
A backup isn't done until it's restored; a fix isn't done until it's tested from the user's side.

Write it down
Every system ships with a runbook someone else could follow; every failure gets a post-mortem.

Direct, then verify
Agents build to my specification, an independent reviewer checks the work, and I verify it before it ships.

Learn by shipping
Linux, Go, AI agents, post-production: each learned by building something real, with continuous self-study.

COMPUTER SCIENCE

From the ground up

Programming	C, C++ with object-oriented design, templates and the STL, Python, functional programming in Standard ML and logic programming in Prolog; recursion, memory and pointers.
Algorithms	Lists, stacks, queues, search trees, heaps, hashing and graphs; divide and conquer, greedy methods, dynamic programming, graph algorithms, network flow and NP-completeness.
Theory	Discrete mathematics, logic and proof, automata and formal languages, Turing machines, decidability and complexity, and compiler front ends: lexing and parsing.
Computer systems	Data representation, x86-64 and MIPS assembly, digital logic design, datapaths and pipelining, caches and the memory hierarchy, and program optimisation.
Operating systems	Linking, processes and signals, virtual memory and allocators, Unix I/O, sockets, threads and synchronisation, scheduling, deadlock and file systems.
Networks	HTTP, DNS, TCP and UDP, congestion control, IP routing and NAT, Ethernet and wireless, network security, and distributed systems: clocks, replication and consistency.
Databases	The relational model, SQL and PL/SQL, ER design, normal forms, transactions, concurrency control and recovery.
Software engineering	Requirements and specification, architecture and design, testing and validation, configuration and release management, Agile delivery, ethics and usability.
AI and security	Search, logic, probabilistic reasoning and learning; cryptography from RSA and Diffie-Hellman to hashes, MACs and TLS; access control and web security.
Mathematics and science	Calculus, linear algebra, probability and statistics in R, numerical methods, physics from mechanics to electricity and magnetism, and engineering graphics.

TOOLKIT

What I work with

Languages	Go, Python, TypeScript and JavaScript, Bash, SQL, C, C++, HTML and CSS, Standard ML, Prolog, R, x86-64 and MIPS assembly.
Front end	Astro, Preact, React, Tailwind CSS and Vite; installable offline web apps, IndexedDB, Server-Sent Events, WebAssembly, pdf.js, Canvas and WebGL.
Back end and data	The Go standard library, FastAPI and Pydantic, SQLite with write-ahead logging and full-text search, JSON APIs and JSON Schema.
Testing and quality	Vitest, Playwright, axe, Lighthouse CI, Go tests with the race detector and fuzzing, pytest, ShellCheck and gitleaks.
Infrastructure	Linux (Debian, Arch, Ubuntu), Docker and Compose, systemd, Caddy, Tailscale, AdGuard Home, Syncthing, Home Assistant, GitHub Actions and Raspberry Pi.
AI and agents	Coding agents with hooks, skills and subagents, MCP tool servers, model APIs and local inference, and image, video, voice and music generation models.
Media	ffmpeg, GPU hardware encoding, EBU R128 loudness, numpy signal processing, colour and audio calibration.
Smaller tools	**agent-guardrails** 36 tests Agents that judge work they did not write, and a warning before a destructive command. **aih** 31 tests Every AI conversation from every machine in one local full-text index. **bwtidy** 12 tests Audits and tidies a password vault, with a plan you approve before anything changes. **filmforge** 40 tests A tested production method for painted films made with AI models.

EDUCATION

Computer Science and Engineering

CONTACT

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Remote, on UTC+3: the full European working day overlaps.

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